CROSS-AD · OPTION 1 PROTOTYPE

Shared EEG–MRI embedding from unpaired cohorts: results

22 October 2026 · code `train_option1.py` (job script `train_option1.qsub`) · outputs `results/option1/`

Bottom line. Training EEG together with unpaired ADNI MRI raised EEG AUC over an EEG-only network (0.691 → 0.725), but the ablations show the gain is not MRI information: shuffling the MRI labels keeps most of it, and replacing MRI with age/sex alone does slightly better than MRI (-0.012). The joint model only matches plain EEG logistic regression (-0.002). The contrastive loss acts as a regularizer for a small network; we find no evidence that unpaired MRI improves EEG prediction (or vice versa).

1. Data

Modality	Cohort	Subjects (AD / CN)	Input features
EEG	BrainLat (2 sites) + OpenNeuro ds004504 (1 site), resting state	71 / 61 (35/32 + 36/29)	26 spectral features: relative δ, θ, α, β power (global + 4 scalp regions), log θ/α ratio (global + regions), peak alpha frequency
MRI	ADNI, baseline 3T, CN age/sex-matched 2:1	294 / 576	321 UCSF FreeSurfer 7 regional measures (volume, area, thickness); volumes ÷ intracranial volume; 2 near-empty columns dropped

No subject has both modalities. EEG was preprocessed identically for both datasets (average reference, 1–30 Hz, 256 Hz, ≤300 s, Welch PSD). Age, sex and education are *not* model inputs because they are confounded with diagnosis in BrainLat. Details: `reports/data_introduction.pdf`.

2. Method

Model

- **Encoders**: one MLP per modality (EEG 26→32→16; MRI 321→64→16; ReLU, dropout 0.3) into a shared 16-d embedding.
- **Shared head**: one linear AD/CN classifier used by both modalities, so "AD" means the same direction in the embedding.
- **Cross-modal supervised contrastive loss** ($\tau = 0.1$): each EEG embedding is pulled towards MRI embeddings with the same label and away from the other label, and vice versa. No subject pairing is used — only labels.
- **Loss** = $BCE_{target} + 1.0 \cdot BCE_{auxiliary} + 0.5 \cdot$contrastive; BCE class-weighted for imbalance.
- Full-batch Adam, lr 0.001, weight decay 0.001, 300 epochs. All hyperparameters fixed before running; nothing was tuned on test folds.
- At test time a subject is predicted **from its own modality only**.

Evaluation

- **EEG CV**: 5-fold × 10 repeats over all EEG subjects, stratified by dataset × diagnosis; all MRI subjects are auxiliary training data.
- **EEG transfer**: train on one EEG dataset (+ MRI), test on the other; 10 seeds.
- **MRI CV**: 5-fold × 3 repeats over ADNI; all EEG subjects auxiliary.
- Imputation and z-scoring fitted on training rows only (EEG per dataset; the unseen transfer dataset is z-scored with its own, label-free statistics).
- Metric: AUC of out-of-fold predictions per repeat; every model uses the same splits, so differences are paired.
- **Comparisons**: logistic regression and the same network without the other modality; ablations without the contrastive loss, with auxiliary labels shuffled, and with auxiliary nodes that carry only age + sex.

3. Results

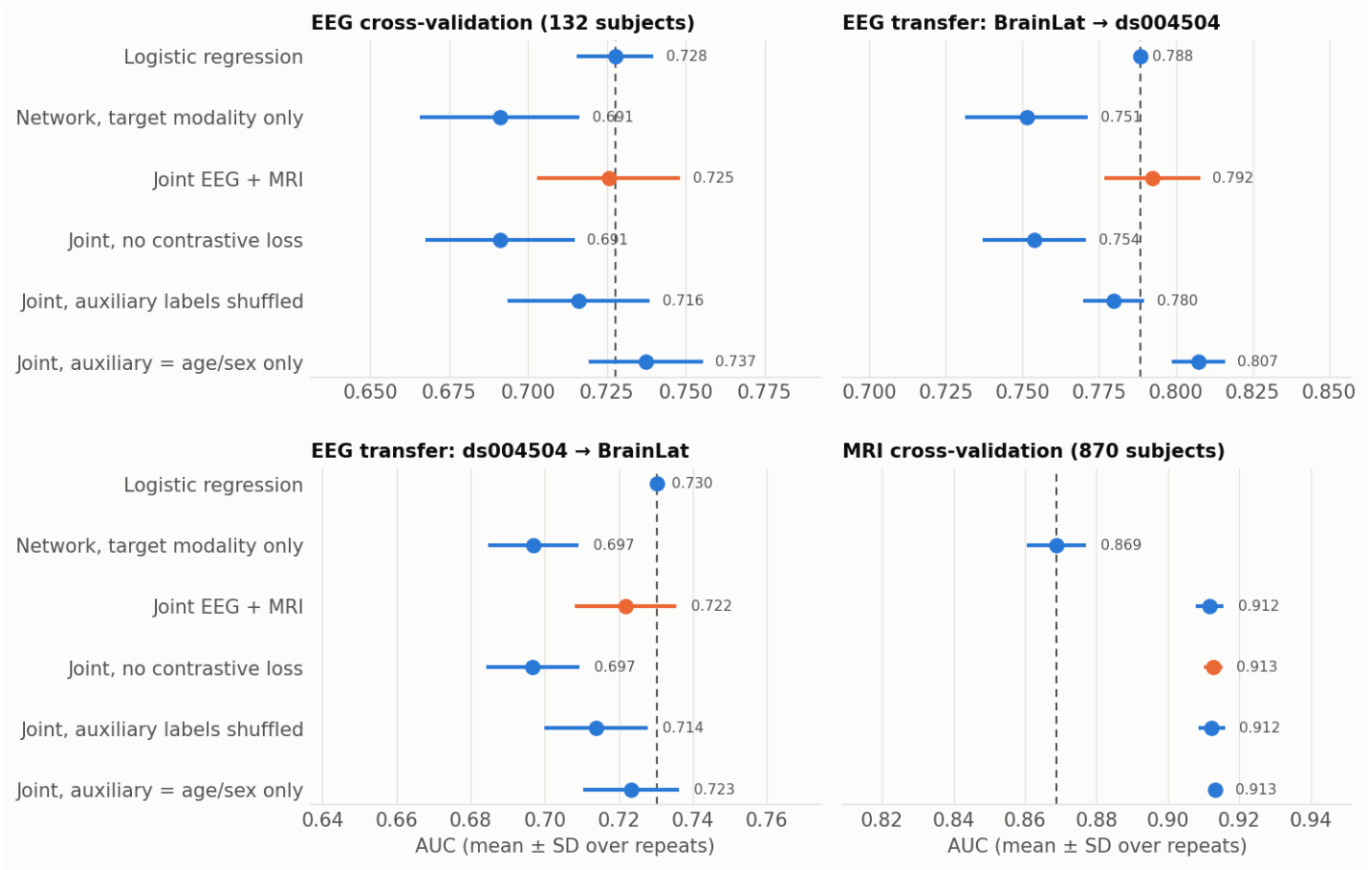


Figure 1. AUC per model (mean ± SD over repeats). Orange = joint EEG + MRI model; dashed line = logistic regression.

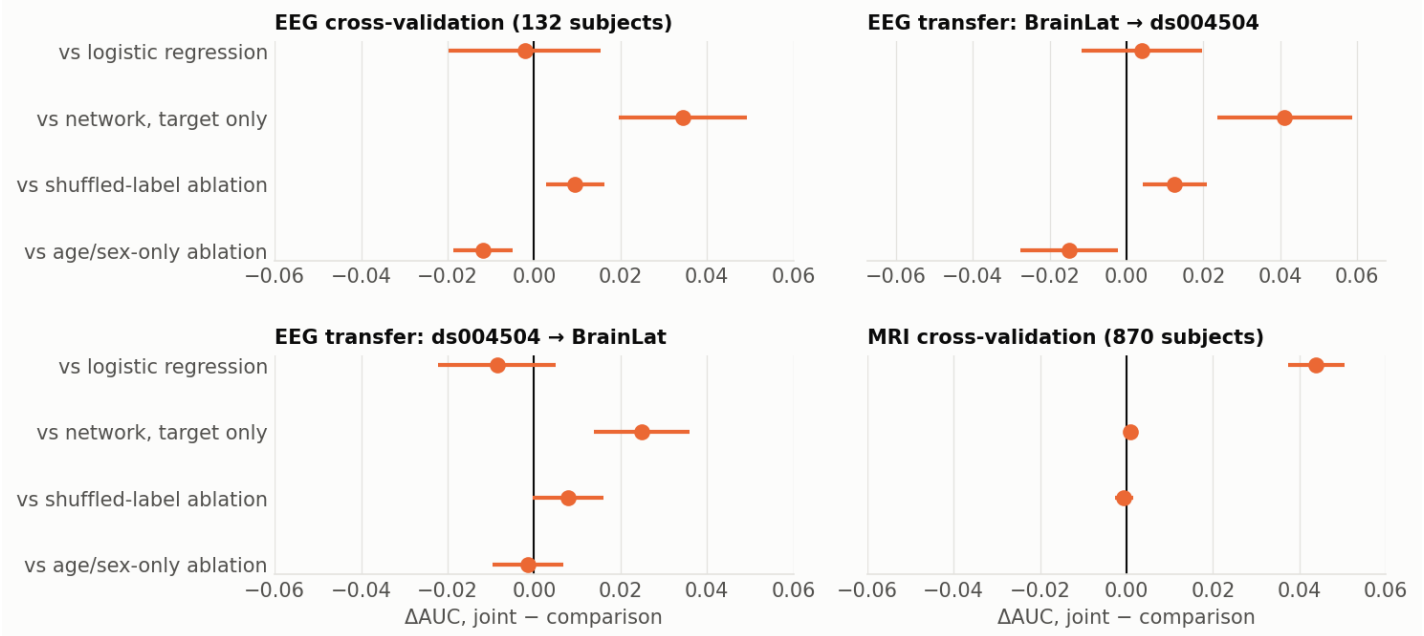


Figure 2. Paired AUC difference of the joint model against each comparison (same splits). Right of 0 = joint better. If MRI carried useful information, the joint model should beat the shuffled and age/sex-only ablations clearly; it does not.

Joint model vs comparisons (paired)

Track	Joint model	ΔAUC (mean ± SD)	Joint better in
EEG cross-validation (132 subjects)	vs logistic regression	-0.002 ± 0.018	3 / 10
EEG cross-validation (132 subjects)	vs network, target only	+0.034 ± 0.015	10 / 10
EEG cross-validation (132 subjects)	vs shuffled-label ablation	+0.009 ± 0.007	9 / 10
EEG cross-validation (132 subjects)	vs age/sex-only ablation	-0.012 ± 0.007	0 / 10
EEG transfer: BrainLat → ds004504	vs logistic regression	+0.004 ± 0.016	4 / 10
EEG transfer: BrainLat → ds004504	vs network, target only	+0.041 ± 0.018	10 / 10
EEG transfer: BrainLat → ds004504	vs shuffled-label ablation	+0.013 ± 0.008	10 / 10
EEG transfer: BrainLat → ds004504	vs age/sex-only ablation	-0.015 ± 0.013	1 / 10
EEG transfer: ds004504 → BrainLat	vs logistic regression	-0.009 ± 0.014	2 / 10
EEG transfer: ds004504 → BrainLat	vs network, target only	+0.025 ± 0.011	10 / 10
EEG transfer: ds004504 → BrainLat	vs shuffled-label ablation	+0.008 ± 0.008	7 / 10
EEG transfer: ds004504 → BrainLat	vs age/sex-only ablation	-0.002 ± 0.008	4 / 10
MRI cross-validation (870 subjects)	vs logistic regression	+0.044 ± 0.007	3 / 3
MRI cross-validation (870 subjects)	vs network, target only	+0.001 ± 0.001	2 / 3
MRI cross-validation (870 subjects)	vs shuffled-label ablation	-0.001 ± 0.002	1 / 3

Per-model AUC for every track is in Figure 1 and `results/option1/summary.csv`.

EEG CV AUC within each EEG dataset

subset	BrainLat	ds004504
Model		
Logistic regression	0.653	0.798
Network, target modality only	0.613	0.774
Joint EEG + MRI	0.645	0.804
Joint, no contrastive loss	0.612	0.774
Joint, auxiliary labels shuffled	0.634	0.796
Joint, auxiliary = age/sex only	0.661	0.808

4. Interpretation

- **EEG** separates AD from CN with AUC ≈ 0.73 overall (≈0.80 within ds004504, ≈0.65 within BrainLat) and transfers across datasets at 0.73–0.79.
- **MRI** regional features reach AUC ≈ 0.91; adding EEG as auxiliary data changes nothing (+0.001).
- The joint model's gain over the EEG-only network (+0.034) disappears against the ablations: the shared head alone adds 0.000, the shuffled-label ablation keeps most of the gain, and age/sex-only auxiliary nodes do better than MRI. The contrastive term regularizes the small EEG network, which otherwise overfits; logistic regression gets the same benefit for free.
- This is the controlled negative result anticipated in the proposal: with unpaired cohorts, the label-aligned shared embedding does not transfer disease information between modalities beyond regularization.

5. Limitations and next steps

- Hyperparameters were fixed, not tuned (nested CV would tune them without test leakage); the EEG-only network had no early stopping, so logistic regression is the fairer baseline.
- 132 EEG subjects: AUC differences of ≈0.01 are within noise (SD across repeats 0.01–0.03).
- BrainLat confounds (recording type matches diagnosis, AD older and less educated) limit what the BrainLat numbers mean.
- Next: nested-CV tuning, EEG connectivity features, global-only EEG features for cross-dataset robustness, feature importance; and paired EEG–MRI fusion once BrainLat MRI download access is granted.